```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.metrics import accuracy_score

iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.3, random_state=42)

model = DecisionTreeClassifier(criterion='entropy', max_depth=3, random_state=42)
model.fit(X_train, y_train)

pred = model.predict(X_test)
print("Accuracy: %.4f" % accuracy_score(y_test, pred))
print("\nDecision tree rules:")
print(export_text(model, feature_names=list(iris.feature_names)))
sample = [[5.1, 3.5, 1.4, 0.2]]
print("Prediction for", sample, "->", iris.target_names[model.predict(sample)[0]])
```

```
Accuracy: 0.9778

Decision tree rules:
|--- petal length (cm) <= 2.45
|   |--- class: 0
|--- petal length (cm) >  2.45
|   |--- petal length (cm) <= 4.75
|   |   |--- petal width (cm) <= 1.60
|   |   |   |--- class: 1
|   |   |--- petal width (cm) >  1.60
|   |   |   |--- class: 2
|   |--- petal length (cm) >  4.75
|   |   |--- petal length (cm) <= 5.15
|   |   |   |--- class: 2
|   |   |--- petal length (cm) >  5.15
|   |   |   |--- class: 2

Prediction for [[5.1, 3.5, 1.4, 0.2]] -> setosa
```